Chapter-1st: Electricity and Magnetism

Electric charge: The fundamental property of fundamental particles that constitute matter is called charge.

Q:1 What is charge?

Q:2 What is quantization of charge?

Answer: Any amount of charge cannot exist in a body. If 'e' is the charge of an electron, then total charge in a body $q = ne$; where n is a positive or negative integer. Subsequent experiments show that electric charge is not continuous but is made up of a certain minimum electric charge. This fundamental charge is the charge of an electron or proton and has the magnitude 1.60210×10^{-19} Coulomb.

For example, the 1-coulomb charge is equal to the charge of 6.24×10^{28} electrons.

Q:3 what is conservation of charge?

Answer: A principle in physics: the total electric charge of an isolated system remains constant irrespective of whatever internal changes may take place.

Ex: A glass rod rubbed with a piece of silk cloth gets positively charged, where as the piece of silk cloth become negatively charged.

Conductor and insulator:

Conductor: A conductor, or electrical conductor, is a substance or meterial that allows electricity to flow through it. In a conductor, electrical charge carriers, usually electrons or ions, move easily from atom to atom when voltage is applied.

Insulator: An electrical insulator is a material in which electric current does not flow freely. The atoms of the insulator have tightly bound electrons which cannot readily move. Other materials - semiconductors and conductors - conduct electric current more easily.

Q:4 Explain conductor and insulator in the eye of band theory.

Answer:

In conductor, the valence band is partially filled and it overlaps with a high energy unoccupied conduction band. Electrons can easily flow under an applied electric field. In case of insulator, the valence band is fully-filled and there is large energy gap between the valence band and the conduction band.

Electric Field :

If a charged object is brought near to an electrified body it experiences attractive or repulsive force. The region around the electrified body where its influence exists is called its electric field.

The region around the electric charge in which the electric forces act,

$$E = \frac{F}{q_0}$$

$$= \frac{\frac{1}{4\pi\epsilon_0} \frac{q_1 q_0}{r^2}}{q_0}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_0}{r^2} \times \frac{1}{q_0}$$

Colomb's law: At a particular medium the magnitude of attractive or repulsive force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of distance between the charges. The force acts along the straight line joining the two charges.

Explanation:

Let q_1 and q_2 be two point charges and r be the distance between them. If the magnitude of force is F, then according to coulomb's law —

$$F \propto q_1 q_2$$

and $F \propto \frac{1}{r^2}$

$$\therefore F \propto \frac{q_1 q_2}{r^2}$$

$$\text{or } F = K \frac{q_1 q_2}{r^2}$$

$$\text{or, } F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$\therefore F = 9 \times 10^9 \frac{q_1 q_2}{r^2}$$

$M_e = 6\times10^{24}$ kg $= 6000 \times 10^7\,10^7\,10^7$

$M_s = 2\times10^{30}$ kg $= 200 \times 10^7\,10^7\,10^7\,10^7$

Newton's Law: f is the magnitude of the force acting on either particle owing to the charge on the other, and q_1 and q_2 are the magnitudes of the charges of the two particles. The gravitational force between the two particles is given by

$$f = G\frac{m_1 m_2}{r^2}$$

Q:5 what is Coulomb's law of electric field ? what is the difference between Coulomb's law and Newton's law ?

Answer:

Based on the series of experiments by charles Coulomb's law 1785 states that:

the force of electrostatic interaction between two point charges is proportional to the multiplication of their charge modules and reversely proportional to the square of distance between them.

<u>Difference between</u>

In the case of coulomb's law, the forces are inversely proportional to the square of the distance between the charges. and in the case of Newton's law of gravitation the forces are inversely propotion to the square of the distance between the masses.

- $F = K Q_1 Q_2 / r^2$ ⟶ coulomb's law
- $F = G M_1 M_2 / r^2$ ⟶ Newton's law
- Both are inverse square laws $F \propto \frac{1}{r^2}$
- Both have a proportionally to a product of each body mass for gravity electric

charge for electricity.

- A major difference is that gravity is always an attractive force whereas the electric force can be either attractive or repulsive.
- Electrical force is stronger than gravitational force.

Electric potential:

The amount of work done in bringing a unit positive charges from infinity to a point in electric field is called electric potential.

$$V = Ed = \frac{W}{q}$$

Explanation:

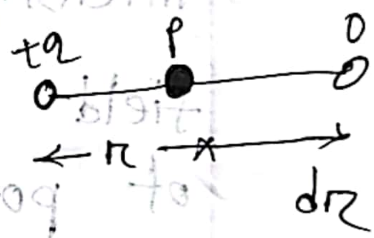
The amount of work done in bringing a unit positive charge from infinity to a point in an electric field.

$$V = \frac{w}{q} = JC^{-1}$$

Electrical potential due to point charge:

Let P be a point at a distance r from a point charge q placed at O. Let the permittively constant of the medium be ε_0. The intensity at P is given by,

$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2} \text{ ——— (i)}$$



The work done to displace a unit charge through

$$-dV = -\frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} dr \text{ ——— (2)}$$

the negative sign indicates decrease of potential

Hence,

$$V = \int_{\infty}^{r} -\frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} dr$$

$$= -\frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r}\right]_{\infty}^{r}$$

$$= -\frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left[-\frac{1}{4\pi\epsilon_0}\frac{q}{r}\right]$$

$$= +\frac{q}{4\pi\epsilon_0}\frac{d}{dr}\left(\frac{1}{r}\right)$$

$$= \frac{1}{4\pi\epsilon_0}\cdot\frac{q}{r^2}$$

$$= E$$

intensity at a point in an electronic field is equal to the negative gradient of potential at same point.

Q:6 Define electric field and electric potential?

Q:7 what is electric field strength?

Answer: Electric field strength is a quantitative expression of the intensity of an electric field at a particular location. The standard unit is the volt per meter (V/m or Vm^{-1}). A field strength of 1 V/m represents a potential difference of one volt between points sepresents ~~a potential di~~ by one meter.

MATH

① Two protons in an iron nucleus are at a distance of 4×10^{-15} m. Find the electric force acting between.

Answer:

$$f = 9 \times 10^9 \frac{q_1 q_2}{r^2}$$

$$= 9 \times 10^9 \frac{q^2}{r^2}$$

$$= 9 \times 10^9 \frac{(1.6 \times 10^{-19})^2}{(4 \times 10^{-15})^2}$$

$$= 14.4 \text{ N}$$

$q_1 = 1.6 \times 10^{-19}$ c $= q_2$

$r = 4 \times 10^{-15}$ m

$f = ?$

(2) Two equally charged balls are separated by 2 mm in air they repel each other with a force of 4×10^{-5} N. calculate the charge in each ball.

Answer:

$$4\times10^{-5} = \frac{q^2}{(2\times10^{-3})^2} \times 9\times10^9$$

$$q^2 = (4\times10^{-5})(2\times10^{-3})^2 / (9\times10^9)$$

$$\therefore q = 1.33 \times10^{-10} C$$

(3) When a 5×10^{-9} test charge is placed at a point, it experiences a force of 2×10^{-4} N in the x-direction that is electric field at that is electric field at that point.

Answer:

Here,

$F = 2\times10^{-4}$ N

$q = 5\times10^{-9}$ C

we know that,

$$F = qE$$

$$E = \frac{F}{q}$$

$$= \frac{2 \times 10^{-4}}{5 \times 10^{-9}}$$

$$= 4 \times 10^{4} \ NC^{-1}$$

4. What is the force on an electron placed at the point where the electric field intensity is $4 \times 10^{-4} \ NC^{-1}$

Answer:

Here,

$$E = 4 \times 10^{-4} \ (NC^{-1})$$

$$q = 1.6 \times 10^{-19} \ C \quad (\text{electron})$$

we know that,

$$F = qE$$

$$= (1.6 \times 10^{-19})(4 \times 10^{-4})$$

$$= 6.4 \times 10^{-23} \ N$$

(5) Calculate the field E due to a dipole moment 4.5×10^{-10} cm^{-1}. at a distance 1 m from it on the perpendicular bisector.

Answer: Given that,

dipole moment, $p = 4.5 \times 10^{-10}$ C/m

$r = 1$ m

We know that

$$E = \frac{p}{4\pi\epsilon_0} \cdot \frac{1}{r^3}$$

$$= \cancel{(9 \times 10)} \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3}$$

$$= (9 \times 10^9) \times \frac{(4.5 \times 10^{-10})}{(1)^3}$$

$$= 4.05 \text{ V/m}$$

(6) What is the potential at the surface of an Aluminium nucleus $r = 4.5 \times 10^{-13}$ cm and it's charge $q = 13$ C.

Answer:

Given that,

$$r = 4.5 \times 10^{-15}\ \text{cm}$$

$$q = 13 \times (-1.6 \times 10^{-19})$$

$$= -2.08 \times 10^{-18}\ C$$

potential, $V = ?$

charge density, $\sigma = \frac{\phi}{A}$

$$= \frac{\phi}{4\pi r^2}$$

we know that,

$$E = \frac{V}{r} \quad \text{---(i)}$$

$$E = \frac{\sigma}{\epsilon_0} \quad \text{---(ii)}$$

Equating (i) and (ii) =)

$$\frac{V}{r} = \frac{\sigma}{\epsilon_0}$$

or, $\frac{V}{r} = \frac{q}{\epsilon_0 4\pi r^2}$

or, $V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$

or, $V = (9 \times 10^9) \times \frac{2.08 \times 10^{-18}}{4.5 \times 10^{-15}}$

$$= 4.16 \times 10^6\ V$$

Gauss's Law : Gauss's law state that the flux of electric field through any closed surface is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed by the surface. If the surface does not enclosed any charge the flux of $\vec{E}$ is zero.

The flux of electric field $\vec{E}$ on a small area of surface = ds

Amount of charge = q

when q charge is inside the surface that means,

$$\oint \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$$

when q charge is outside the surface that means surface that means surface does not close.

$$\oint \vec{E} \cdot \vec{ds} = 0.$$

Columb's law from gauss's law :

Let us consider a sphere of radius r and q amount of charge is enclosed by the surface area of the surface, $S = 4\pi r^2$

Again, let small area $\vec{ds}$ and flux of electric field $\vec{E}$ passing parallely with $\vec{ds}$ through the enclosed surface that means the angle between ds and $\vec{E}$ is zero.

$$\vec{E} \cdot \vec{ds} = Ed\cos\theta$$

$$= Eds$$

According to Gauss's law,

$$\oint \vec{E} \cdot \vec{ds} = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E \oint ds = \frac{q}{\varepsilon_0}$$

$$\Rightarrow ES = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E \times 4\pi r^2 = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

Let us consider a test charge q_0 in the enclosed surface and force between q and q_0 is

$$E = \frac{F}{q_0}$$

$$\therefore F = qE$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{qq_0}{r^2}$$

This is the from of columb's law.

Q:8 State and explain Gauss's law. Mention it's application ?

Answer: Gauss's law states that the total electric flux out of a closed surface is equal to the charge exclosed divided by the permittivity. The electric flux in an area is defined as the electric field multiplied by the area

CamScanner

of the surface projected in a plane and prependicular to the field.

Gauss's law for electricity states that the electric flux ϕ across any closed surface is proportional to the net electric charge q enclosed by the surface that is $\phi = q/\varepsilon_0$, where ε_0 is the electric permitivity of free space and has a value of 8.854×10^{-12} square coulombs per newton per square metre.

Q:9 What is electric dipole ?

Answer: An electric diople is defined as a couple of opposite charges q and -q separated by a distance. By default the direction of electric dipoles in space is always from negative charge -q to positive charge q. Themidpoint q and -q is called the centre of the dipole.

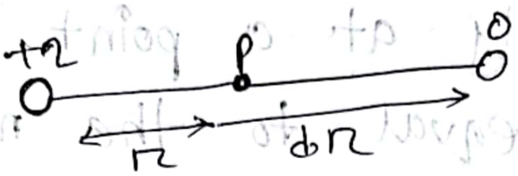
Q:10 what is dielectric in Gauss's law?

Answer: A negative charged $-q$ and $+q$ or the dielectric are called the induced chargeds or bound charges while the charges $+q$ and $-q$ on the capacitor plates are called free charges These induced charges produce their own field which opposes the external field E_0. Let E be the resultant field within the dielectric.

Potential due to a point charge:

Let P be a point at a distance r from a point charge q placed at O. Let the permititively constant of the medium be ε_0. The intensity at P is gives by

$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2} \quad (1)$$



The work done to displace a unit charge through dr

$$-dV = -\frac{1}{4\pi\epsilon_0}\cdot\frac{q}{r^2}dr \quad \text{—— (ii)}$$

The negative sign indicates decreases of potential

$$V = \int_{\infty}^{r} -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}dr$$

$$= -\frac{q}{4\pi\epsilon_0}\left[-\frac{1}{r}\right]_{\infty}^{r}$$

$$= \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left[\frac{1}{4\pi\epsilon_0}\frac{q}{r}\right]$$

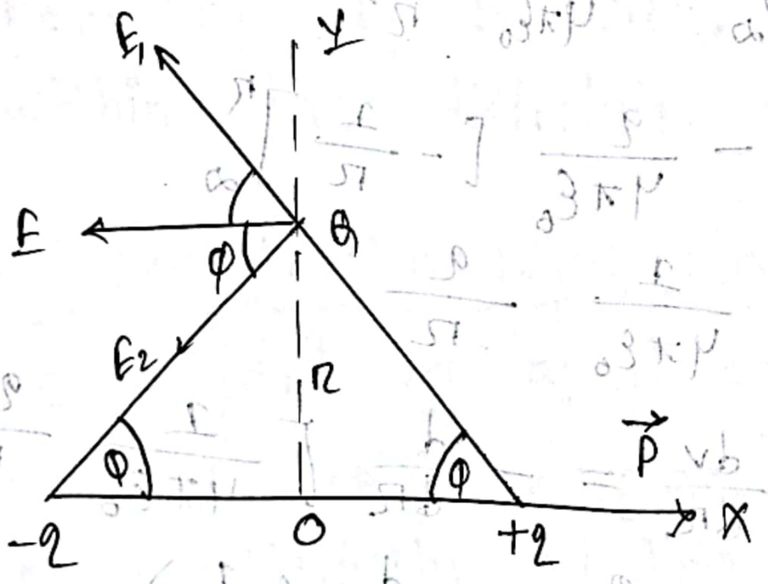
$$= -\frac{q}{4\pi\epsilon_0}\cdot\frac{d}{dr}\left(\frac{1}{r}\right)$$

$$= \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r^2}\right)$$

$$= \frac{q}{4\pi\epsilon_0}\cdot\frac{1}{r^2} = E$$

i.e. intensity at a point in an electric field is equal to the negative gradiant of potential at the same point.

Electric field due to dipole at a point along perpendicular bisector / on dipole at a point P distance r along with perpendicular bisector:



when two equal on opposite charge +q and -q are placed is such way that distance between them is very small and then they from a dipole.

Dipole moment, $p = 2aq$

For $+q$ charge electric field $= E_1$

$-q$ charge electric field $= E_2$

Resultant electric field $E = E_1 + E_2$

$$E_1 = E_2 = \frac{1}{4\pi\varepsilon_0} \frac{q}{(\sqrt{d^2+r^2})^2}$$

$$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{d^2+r^2}$$

From the figure,

$$E = \sqrt{E_1^2 + E_2^2 + 2E_1E_2\cos 2\theta}$$

$$= \sqrt{2E_1^2 + 2E_1^2\cos 2\theta}$$

$$= \sqrt{2E_1^2(1+\cos 2\theta)}$$

$$= \sqrt{2E_1^2 \cdot 2\cos^2\theta}$$

MATH

(7) Two equal charges of magnitude 2×10^{-6} C are placed at a distance 8 cm from each other. Find the magnitude of electric intensity at point 3 cm from the mid point of the line to the two charges along the perpendicular bisector of the line.

Solution:

We know that,

$$E = \frac{2aq}{4\pi\varepsilon_0} \times \frac{1}{(a^2+r^2)^{3/2}}$$

$$= \frac{1}{4\pi\varepsilon_0} \times \frac{2aq}{(a^2+r^2)^{3/2}}$$

$$= (9\times10^9) \times \frac{0.8\times(2\times10^{-6})}{\{(0.4)^2+(0.3)^2\}^{3/2}}$$

$$= 9\times10^9 \times 1.28\times10^{-5}$$

$$= 1.152\times10^5 \text{ N/C}.$$

Given that,

$q = 2\times10^{-6}$ C

$2a = 8$ cm $= 0.8$ m

$a = 4$ cm $= 0.4$ m

$r = 3$ cm $= 0.3$ m

$\frac{1}{4\pi\varepsilon_0} = 9\times10^9$

$E = ?$

(8) An electric dipole consists of two opposite charges of magnitude $q = 2\times10^{-6}$ C separate by 1 cm. The dipole is placed in an external field 2×10^{5} N/C calculate.

(i) The maximum torque on the dipole.

(ii) Work done to turn the dipole through 180° starting from a position of $\theta = 0°$

Solution:

(i)

We know that,

$$\tau = pE\sin\theta$$

$$= 2\times10^{-12}\times2\times10^{5}\times\sin 90°$$

$$= 4\times10^{-4}\ \text{Nm}$$

Given that,

$$p = qr = 2\times10^{-6}\times0.01$$

$$= 2\times10^{-8}\ \text{m}$$

$$E = 2\times10^{5}\ \text{N/C}$$

Again,

$$p = 2\times10^{-7}$$

$$E = 2\times10^{5}\ \text{N/C}$$

$$w = ?$$

(ii)

We know that,

$$w = \int_0^{180} PE\sin\theta \, d\theta$$

$$= -PE\left[\cos\theta\right]_0^{180}$$

$$= -PE\,(\cos 180 - \cos 0)$$

$$= -PE\,(-2)$$

$$= 2PE$$

$$= 2 \times 2\times10^{-7} \times 2\times10^{5}$$

$$= 8\times10^{-2}\ \text{J}$$

(9) A 5 columbs charge is placed on the circumference of a circle of radius 0.5 m. Find the potential and intensity at a distance 25 cm from the center.

solution:

(1)

we know that,

$$V = \frac{1}{4\pi\varepsilon_0} \times \frac{q}{r}$$

$$= (9\times10^9) \times \frac{5}{0.5}$$

$$= 9\times10^9 \, V$$

Given that,

$q = 5c$

$r = 0.5m$

potential $V = ?$

$q = 5c$

$r = 25\times10^{-2}$

$E = ?$

Electric field intensity,

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{q}{r^2}$$

$$= (9\times10^9) \times \frac{5}{(25\times10^{-2})^2}$$

$$= 7.2\times10^{11} \, N/C$$

(10) what is the electric potential at the surface of a gold nucleus. The radius is 6.6×10^{-15} m and the atomic number is 79.

Solution:

we know that,

$$V = \frac{1}{4\pi\varepsilon_0} \times \frac{q}{r}$$

$$= (9\times10^9) \times \frac{1.264\times10^{-27}}{6.6\times10^{-15}}$$

$$= 1.723\times10^{-3} \, V$$

Given that,

$r = 6.6\times10^{-15}$

$z = 79$

charge $= q = ze$

$= 79\times1.6\times10^{-19}$

$= 1.264\times10^{-17} c$

$V = ?$

(11) At once time the positive charge in the atom was thought to be distributed unifomly thoughtout a sphere the with a radius of about 1×10^{-10} m. That is throughtout the enfire atom. Calculate the electric field strength at the surface of a told atom. ($z = 79$) on this suption.

Solution:

Given that,

$r = 1\times10^{-10}$ m

$z = 79$

$q = ze$

$= 1.264\times10^{-17}$ C

$E = ?$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

$$= (9\times10^9)\times\frac{1.264\times10^{-17}}{(1\times10^{-10})^2}$$

$$E = 1.1376\times10^{13}\ N/C$$

(12) A negative point charge of 10^{-6}c is distributed in air of the orgin of a rectengutar co-ordinate system. A second negative point charge of 10^{-4}c is situated on the positive x-axis at a distance of

50 cm from the origin. what is the force on the second charge.

<u>Solution:</u>

$$F = \frac{1}{4\pi\varepsilon_0} \times \frac{q_1 q_2}{r^2} \times \hat{r}$$

$$= (9\times10^9) \times \frac{(-10^{-6})(-10^{-4})}{(50\times10^{-2})^2}$$

$$= 3.6\,N$$

$q_1 = -10^{-6}\,C$

$q_2 = -10^{-4}\,C$

$r = 50\times10^{-2}\,m$

$F = ?$

$$\vec{F} = 3.6\,N$$